

What have we learned from Models?

How do we find them useful?

Let me count the ways.

- 1 Weather forecasts
- 2 Estimating state of the ocean.
- 3 Quantitative estimates of global warming.
- 4 El Nino simulation and prediction. Climate forecasts generally.
- 5 etc etc etc

How do we find them fundamental:

- 1 Show some theories are wrong: numerical models are like lab experiments.
- 2 Unpredictability of weather (chaos theory, with caveats)
- 3 Some modern ideas about ocean circulation ('paradigm' shifts of how the MOC works).
- 4 More?